Database Normalization

Week 4 notes on removing redundancy from relational tables, with one worked example carried from first to third normal form.

1 Why normalize

A table that repeats the same fact in many rows invites mistakes. Normalization splits it into smaller tables so that each fact is stored once.[1]

1.1 Update anomalies

Three problems appear when facts are repeated:

- Update anomaly: changing one fact means editing many rows.
- Insert anomaly: a new fact cannot be stored until an unrelated one exists.
- Delete anomaly: removing one row also removes a fact we wanted to keep.

1 E. F. Codd introduced the first three normal forms in 1970 and 1971.

2 First normal form

A table is in first normal form (1NF) when every cell holds a single value and there are no repeating groups of columns.

2.1 Example

Table 1: Enrollment before normalization.

StudentID	Name	Courses
101	Ayesha	DB, OOP
102	Bilal	DB
103	Hina	OOP, DS, DB

The Courses column holds lists, so the table is not in 1NF. Giving each course its own row fixes it:

StudentID	Name	Course
101	Ayesha	DB
101	Ayesha	OOP
102	Bilal	DB
103	Hina	OOP
103	Hina	DS
103	Hina	DB

3 Second normal form

A table is in second normal form (2NF) when it is in 1NF and every non-key column depends on the whole primary key, not on part of it. In the table above the key is (StudentID, Course), but Name depends on StudentID alone.

To reach 2NF:

1. Find every non-key column that depends on part of the key.
2. Move those columns into a new table keyed by that part.
3. Keep the original table with the full key and the remaining columns.

3.1 Result

Table	Columns	Primary key
Student	StudentID, Name	StudentID
Enrollment	StudentID, Course	StudentID, Course

4 Third normal form

A table is in third normal form (3NF) when it is in 2NF and no non-key column depends on another non-key column. Such a chain is called a transitive dependency.

4.1 Transitive dependencies

Suppose Student also stores DeptID and DeptName. DeptName depends on DeptID, which depends on StudentID, so DeptName depends on the key only through DeptID. Move DeptID and DeptName into a Department table and keep DeptID in Student as a foreign key.

4.2 Checklist

- Every cell holds one value (1NF).
- No column depends on part of the key (2NF).
- No column depends on another non-key column (3NF).

5 Summary

Each normal form removes one kind of repetition. Most course databases stop at 3NF, which is enough to prevent the three anomalies listed in section 1.1.